



BRAIN MODELING FOR MEDICOS

A 7-day hands on program on computational Neuroscience, specifically designed for medicos! Moving from Neurons to Disease Models, explore how AI and ML can be used to both understand, predict and one day treat diseases.

JULY 8-14, 2026 (ONLINE)

DAY 01

Why Brain Modeling Matters in Medicine

- Limits of descriptive neuroscience (the "black box" problem)
- Brain as a dynamical system

DAY 02

From Biology to Equations

- Neuron structure: dendrites, axon, synapse
- Ion channels, neurotransmitters, receptors
- Membrane potential, action potential, PSPs

DAY 03

Mathematical Foundations for Brain Modeling

- Vectors, matrices (state representation)
- Regression & classification
- Decision boundaries
- Gradient descent (optimization as learning)

DAY 04

From Biophysics to Computation

- Neuron Models - rate-coded, spiking and oscillatory
- Neural Networks - Feedforward, Recurrent

DAY 05

Feedforward network and the Visual Brain

- Convolutional Neural Networks (CNNs)
- Hands-on Session
- Visual Hierarchy and CNN

DAY 06

Recurrent Network and Learning

- Using recurrent networks to model the brain
- Learning rules - Supervised, Unsupervised, Reinforcement Learning

DAY 07

Disease Modeling

- Decision Systems - Basal Ganglia
- BG Model; Using DBS for PD

CLICK HERE TO APPLY!

HIGHLIGHTS

**BRAIN & DISEASE
HANDS-ON CODING
BRAIN DYNAMICS**

FEE DETAILS

1,500 INR
(exclusive of GST)

DATES:

July 4th - Last date
for registration

CONTACT:

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